Proof

The truth table required for proving the associative law is given below.

p	q	r	$p \vee q$	$q \vee r$	$(p \vee q) \vee r$	$p \vee (q \vee r)$
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	T	T
T	F	F	T	F	T	T
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	F	T	T	T
F	F	F	F	F	F	F

Table 12.16

The columns corresponding to $(p \vee q) \vee r$ and $p \vee (q \vee r)$ are identical.

Hence $p \vee (q \vee r) \equiv (p \vee q) \vee r$.

Similarly, (ii) $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$ can be proved.

4. Distributive Laws

(i) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ (ii) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

Proof (i)

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Table 12.17

The columns corresponding to $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are identical. Hence $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

Similarly (ii) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ can be proved.

5. Identity Laws

(i) $p \vee \mathbb{T} \equiv \mathbb{T}$ and $p \vee \mathbb{F} \equiv p$ (ii) $p \wedge \mathbb{T} \equiv p$ and $p \wedge \mathbb{F} \equiv \mathbb{F}$

p	$\mathbb{T}$	$\mathbb{F}$	$p \vee \mathbb{T}$	$p \vee \mathbb{F}$
T	T	F	T	T
F	T	F	T	F

Table12.18

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